Part-A

(Segment-5)

- The design process and importance
- Design process steps
- Design methodologies → Top down and bottom up
- Structured design.
- Modularization
- ~~Data~~ Input design
- Data input method
- Output design
- Designing effective outputs
- Good form design, objectives
- Database design
- Data modeling
- ERD

Process of design:

The design process: & why it is important:

Achieving good design takes more than simply time and talent. It takes clarity around a specific goal, and when a largen is involved, callabration and allignment.

⇒ The design process is very important to non-designers.

⇒ Ultimately the design process provides healthy guidelines to ensure ~~that~~ your design concepts meet user needs.

→ The design process is crucial because it's the disciplined approach that guides creative teams towards a clear and meaningful goal for their design work.

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What does the design process look like?

⇒ No two ~~two~~ creatives have the same exact process and that's also true for design teams and their design processes.

⇒ Teams and individual designers who are focused on UI design will have a different approach than who are doing graphic design or interaction design.

Design process steps: There are five steps in the design process. They are,

1. Empathize
2. Define
3. Ideate
4. Deliver
5. Test

① Empathize – The 1st step in the design process also referred as to "discovery", involves the same initial effort required for any new business challenge. It includes activities like market research, UX research, conducting user interviews and bringing together all of the information.

② Define – It is all about defining problems, goals and boundaries. It include creating user flows, holding meetings to gain alignment on design scope, deliverables, responsibilities and timelines.

③ Ideate – The third step of the design is designers creative process.

④ Deliver – The step of the design process is the turning point when designs begin to feel real because they shared with

others in the form of a prototype or MVP.

⑤ Test ← Test is ~~designe~~ defined as running test on design by user to get further refined and bugfree digital product. There is a continuous cycle of updates and improvements to the design based on feedback from users and customers.

Importance of design process: Establishing a design process improves team collabration, increases the quality and speed of your design work and make it more likely that everyone.

positive benefits of a solid design process:

① set clearer expectations :- "what's happening with the design?"

② Improve teamwork and alignment.

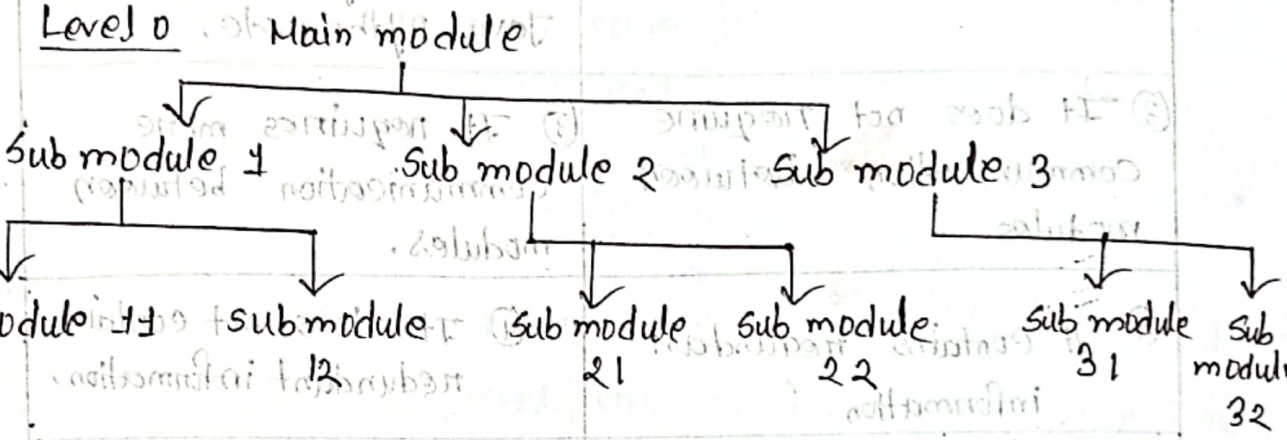
③ create more efficiently - The design process is like a ~~overflow~~ workflow to follow, and we all know that templates help us all work faster.

④ Deliver final designs that meet user and business needs

⑤ Gain happy users - The ultimate goal for any design project is not just to solve a problem but to also make users happy.

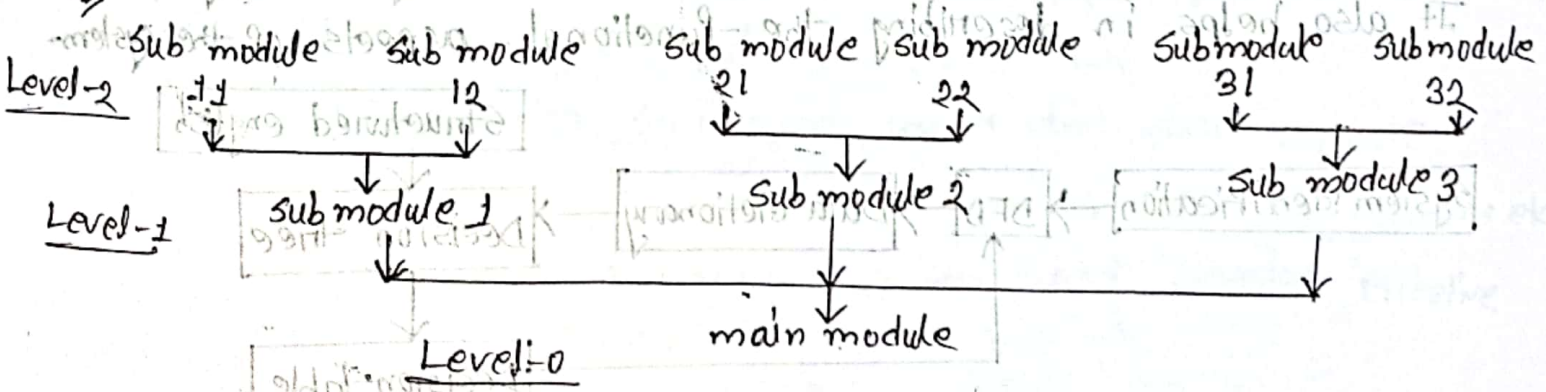
Design methodologies: ① Top down strategy
② Bottom up strategy

① Top down strategy: The top down strategy uses the modular approach to develop the design of a system. It is called so because it starts from the top or the highest level module and moves towards the lowest level modules.

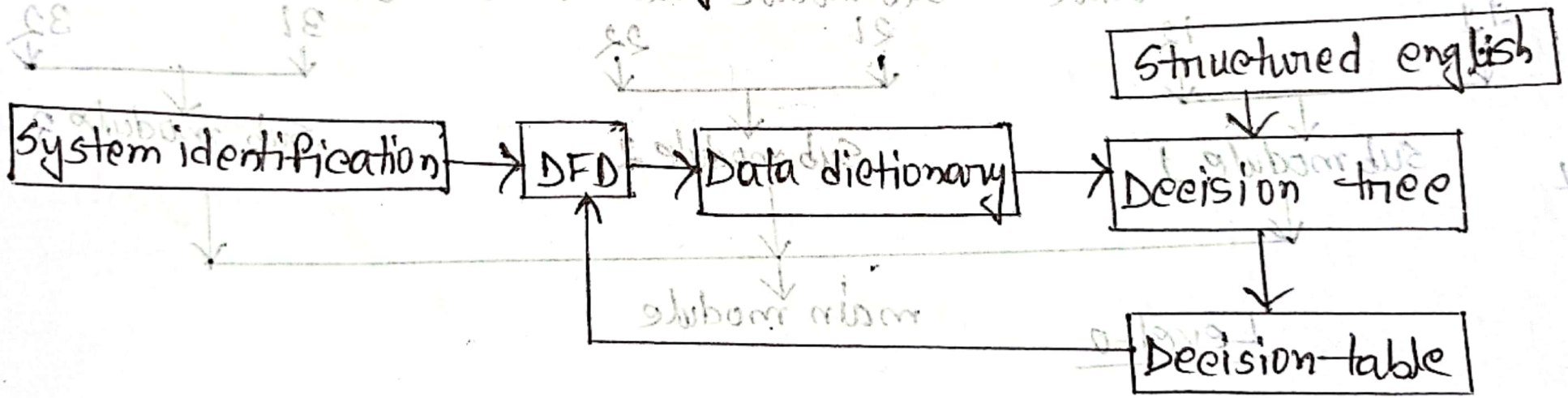


② Bottom up strategy: Bottom up strategy follows the modular approach to develop the design of the system. It is called so because it starts from the bottom or the most basic level modules and moves towards the highest level modules:

→ The modules at the most basic or the lowest level are identified
→ These modules are grouped together
→ These modules are further combined to from the next higher level modules.
→



Structured design: Structured design is a data-flow based methodology that helps in identifying the input and output of the developing system. The main object is to minimize the complexity and increase the modularity of a system. It also helps in describing the functional aspects of the system.



Modularization: Structured design partitions the program into small and independent modules. These are organized in top down manner with the details shown in bottom.

Advantages: ① Critical interfaces are tested first.
② It provide abstraction.
③ It allows multiple programmers to work simultaneously.
④ It allows code reuse.
⑤ It provides control and improves morale.
⑥ It makes identifying structure easier.

Input design: Input design is the raw data that is processed to produce output. The developers must consider the input devices such as PC, MICR, OMR etc.

Properties: ① It should ~~have~~ serve specific such as storing, recordi the information.
② It ensures proper completion with accuracy.
③ It should be easy to fill and straightforward.
④ It should focus on user's attention, simplicity.

Objectives: ① To design data entry and input procedures
② To reduce input volume.
③ To design source documents for data capture
④ To design input data records, data entry screens
⑤ To use validation checks and develop effective input controls.

Data input methods: It is important to design appropriate data input methods to prevent errors while entering data.

- A system should prevent user from making mistakes by -
  1. clear form design by leaving enough space for writing legibly.
  2. clear instructions to fill form.
  3. clear form design
  4. Reducing key strokes.
  5. Immediate error feedback.

- ONLINE: User directly enters data using screen prompts.
- OFFLINE: Forms filled by users.

- Some of the popular data input methods are -
  1. Batch input method (offline data input method)
  2. Online data input method
  3. Computer readable forms
  4. Interactive data input

Output design: The design of the output is the most important task of any system. During output design, developers identify the type of outputs needed and consider the necessary output controls and prototype report layouts.

Objectives:
1. To develop output design that serves the intended purpose and eliminates the production of unwanted output.
2. To develop the output design and meets the user requirements.
3. To deliver the appropriate quantity of output.
4. To form the output of in appropriate format and direct it to the right person.
5. To make the output available on time for making good decisions.

Designing effective outputs: ① Printers ② Display screen ③ Audio output and podcasts ④ DVD, CD-ROM, and CD-RW, ⑤ Electronic output (email, blogs, web sites)

Advantage	Disadvantage
① Printers ⇒ • Affordable • Flexible in types of output • Handles large volume of output	① Printers ⇒ • Expensive supplies • Environmentally unfriendly • Depending on model, may be slow
② Display screen ⇒ • Interactive • Quiet • Good for frequently accessed	② Display screen ⇒ • Require setup space • May require printed documentation
③ Audio output and podcasts ⇒ • Good for individual user • Good for transient message. • Good for where needs hands free	③ Audio output & podcasts ⇒ • Has limited application
④ DVD, CD-ROM, CD-RW ⇒ • Has large capacity • Allows multimedia output	④ • Requires a computer or display for reading data.
⑤ Electronic output ⇒ • Reduces paper • Can be broadcast • Can be updated very easily	⑤ • It is difficult to convey • Websites need diligent maintenance.

Nature of output reports:

- Top management - Graphical output ⇒ pie charts
  ⇒ Bar chards
  ⇒ Maps

Various types of output : ① External outputs
② Internal outputs

- **External outputs:** External output enable the system to leave the trigger actions on the part of their recipients some of external output implemented as a form and re-enter the system a
  ⇒ **Turnaround outputs** are those which are typically implemented as a report eventually re-enters the system as an input

- **Internal outputs:** Internal outputs are present inside the system and used by end-users and managers.
  ⇒ They support the management in decision making & reporting.
  - Detailed reports
  - Summary reports.
  - Exception reports

  } Types of reports

- **Form design:** • Both forms and reports are the product of input & output design.
  • Form provide fields for data input but reports are purely used for reading.

During form design, the designers should know —

⇒ who will use them

⇒ where would they be delivered

⇒ the purpose of the form or report

Examples: Order forms

Objectives of good form design: A good form design is necessary -

① To keep screen simple by giving proper sequence, information & clear captions.

② To meet the intended ~~pers~~ purpose by using appropriate forms.

③ To ensure the completion of form with accuracy

④ To keep form attractive by using icons, inverse video or blinking cursor etc.

⑤ To facilitate navigation.

Good-form design: To design forms that people find useful, four guidelines for form design, should be observed.

① Make forms easy to fill in

② Ensure that forms meet the purpose for which they are designed.

③ Designed forms to ensure accurate completion.

④ keep form attractive

Making forms easy to fill in: Form flow

① Designing a form with proper flow can minimize the time and effort expended by employees in form completion.

② Form should flow form left to right and top to bottom.

③ Illogical flow takes extra time and is frustrating.

④ A form that requires people to go directly to the bottom of the form and then skip back up to the top completion exhibits poor flow.

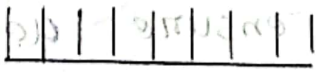
There are 7 ~~main~~ sections of a form to make easy to fill form:

① Heading
② Identification & access
③ Instructions
④ Body
⑤ signature
⑥ Totals
⑦ Comments

Good design and bad design:



Good design



Bad design

Good form design Vs Bad form design:

Good design	Bad design
① Every piece of logic has one & only one home.	① Logic has to be duplicated.
② Nature - simple	② Nature - complex
③ Cost - small	③ Cost - very high
④ The logic link can easily be found.	④ The logic link cannot be remembered
⑤ System can be extended with changes in only one place.	⑤ System cannot be extended so easily.

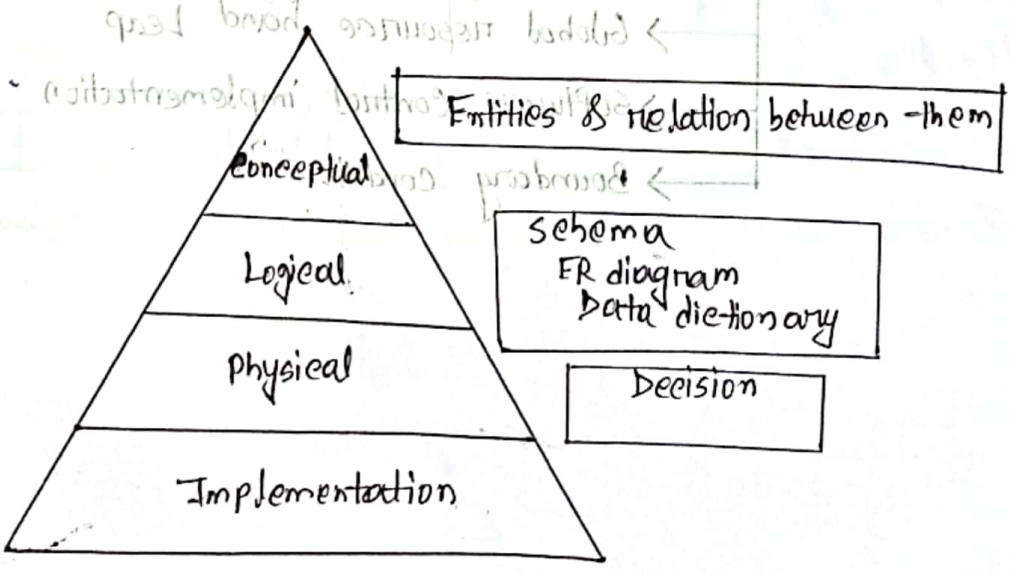
Database design: A database is a collection of information that is organized so that it can be easily accessed, managed and updated. Three main phases in database design, They are, ① conceptual database design
② Logical
③ Physical

⇒ **conceptual database design** ⇒ To build the conceptual representation of database which has the identification of the relationships & attributes.

⇒ **Logical database design** ⇒ To convert the conceptual representation to the logical which includes designing the ERD.

⇒ **Physical database design** ⇒ To decide how the logical structure is to be physically implemented.

Data modeling: Data modeling is often the 1st step in database design as the designers first create a conceptual model of how data items relate to each other.



Conceptual data model: It identifies the highest level relationships between the different entities.

Features: ① Important entities and the relationship among them
② No attribute is specified
③ No primary key is specified.

logical data model: A logical data model is a fully attributed data model that is independent of DBMS technology. It typically describes the data requirements

Features: ① All entities & relationships among them.
② All attributes for each entity are specified.
③ The primary key for each entity is specified.
④ Foreign keys are specified.
⑤ Normalization occurs.

Steps: ① Specify primary keys for all entities
② Find relationship between diffn entities.
③ Find all attributes
④ Resolve many to many relationships
⑤ Normalizations.

Physical data model: It is a fully attributed data model that is dependent upon a specific version of a data technology.

Features: ① Specification all tables & columns.
② Foreign key are used to identify relationship between tables.
③ It will be diffn for RDBMS

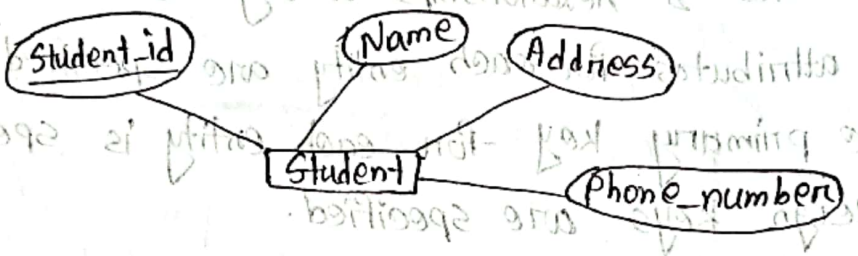
Steps: ① Convert entities into tables.
② Convert relationship into foreign key.
③ Convert attributes into columns.
④ Modify the physical data

Entity Relationship diagram (ERD): Logical representation of entities. Three main constructs are Data entities, Relationships and attributes.

Entity: An entity is a person, place, object, event or concept about which an organization wishes to maintain data.

Attributes: Each entity has a set of attributes. An attribute is a property of an entity.

Example:



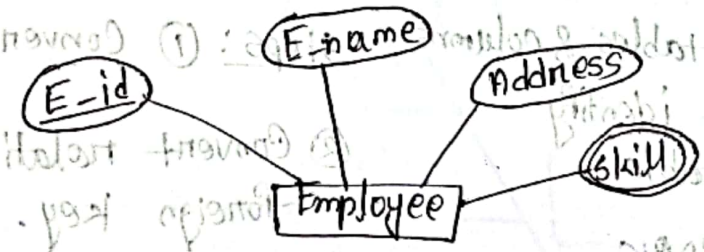
Primary key: A primary key is a column in a table that uniquely identifies the rows in that table

Example: Studen_id

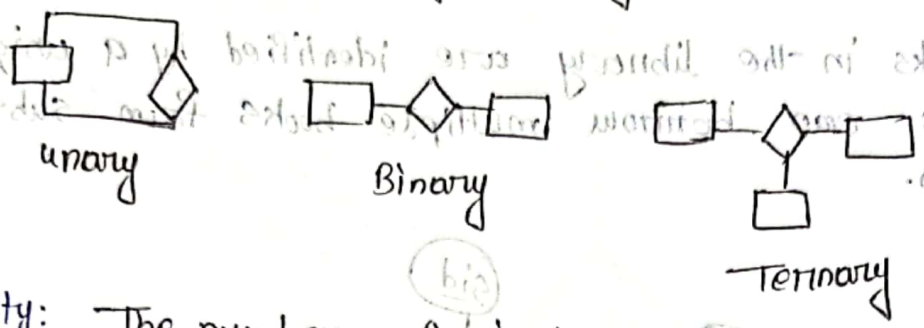
Multivalued attributes: An attribute that may take on more tha one value for each entity instance.

Represented on E-R diagram in two ways, → double lined e
→ weak entity

Example: Double lined ellipse,



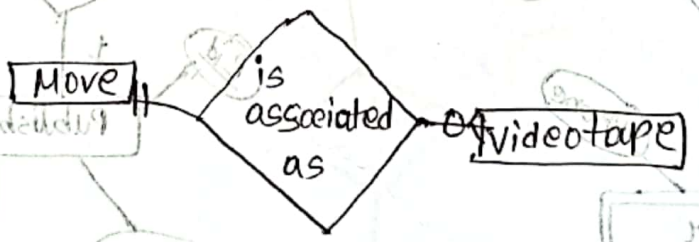
Degree of relationship: Unary ⇒ Between two instances of one entity type
Binary ⇒ Between the instances of two entity types
Ternary ⇒ among the instances of three entity types



Cardinality: The number of instance of entity B that can or must be associated with each instance of entity A.

- **Minimum cardinality:** The minimum number of instance of entity B associated with entity A.
- **Maximum cardinality:** The maximum number of instance of entity B associated with entity A
- **Mandatory vs optional cardinalities:** whether an instance must exist or can be absent in the relationship.

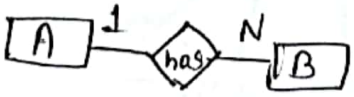
Example:



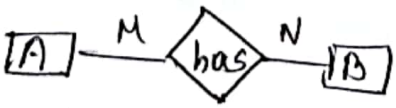
1. One to one :



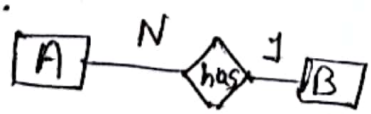
2. One to many :



3. Many to many :



4. Many to one :



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Draw an ER model for library management application.

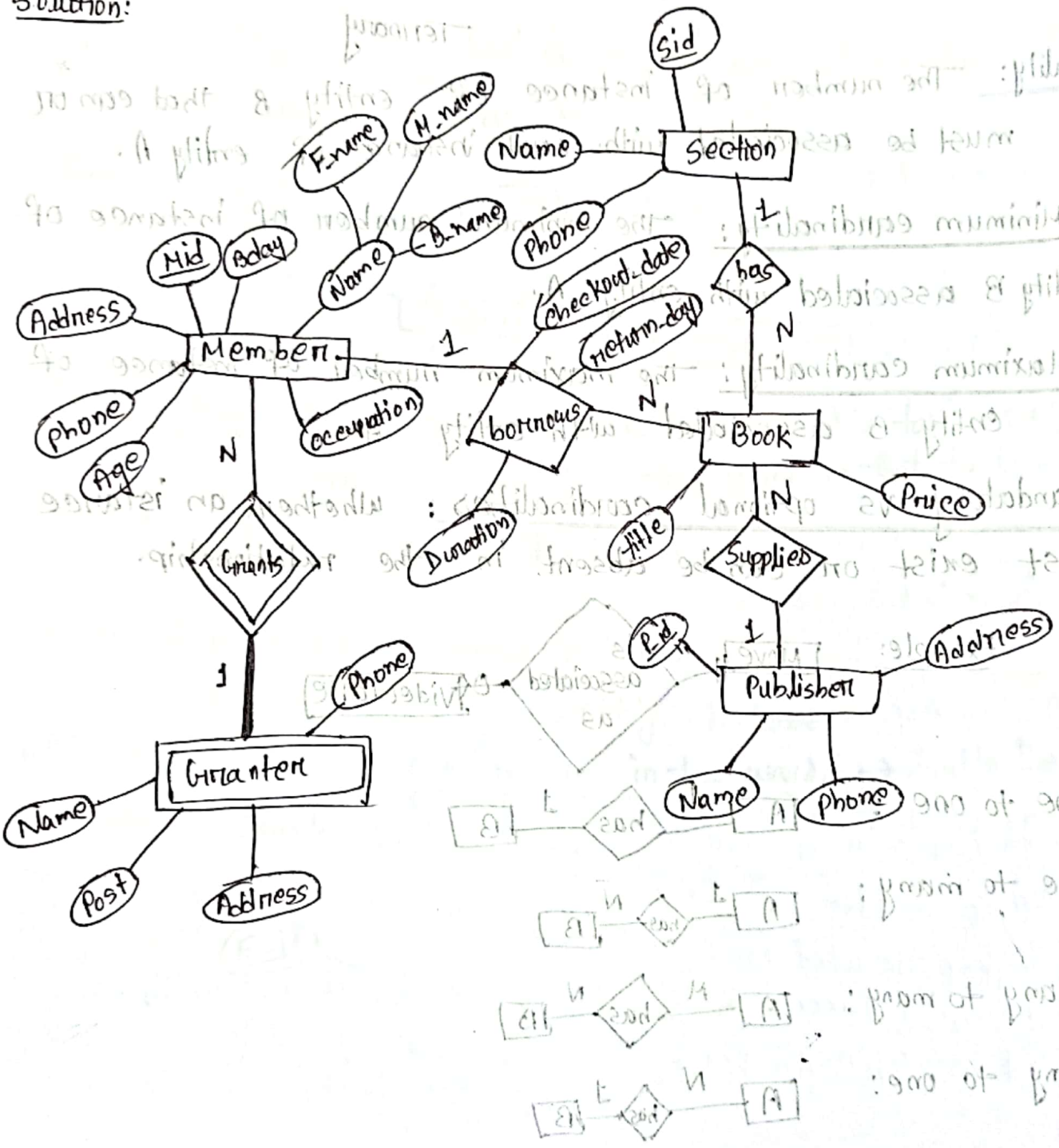
⇒ In a library multiple students can enroll.

⇒ Students can become a member by paying an appropriate fee.

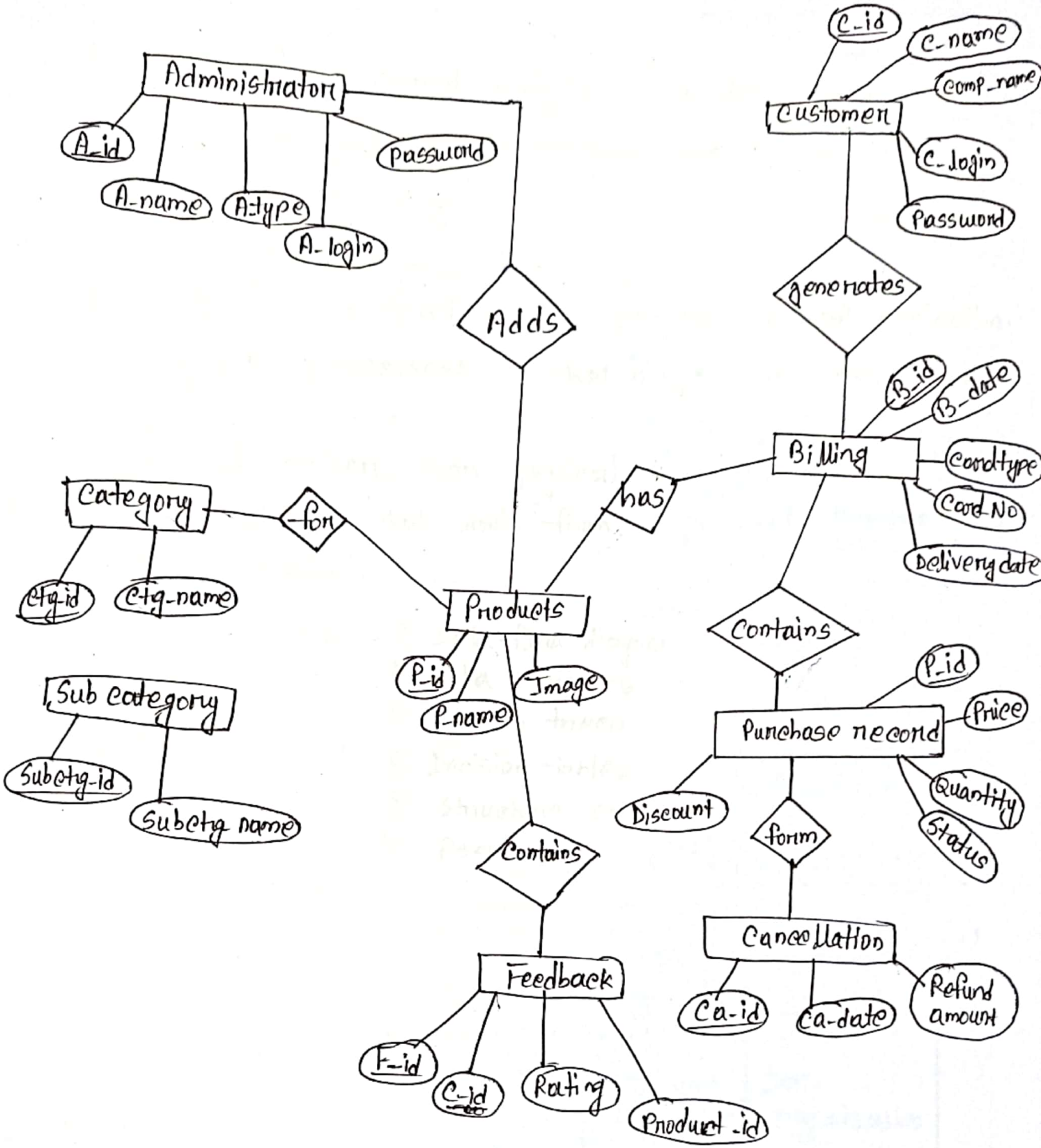
⇒ The books in the library are identified by a unique id.

⇒ Students can borrow multiple books from subscribed libraries.

Solution:



E-commerce Site ERD



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